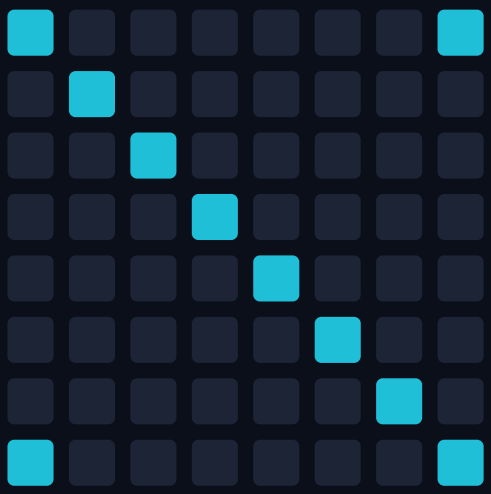
MTGPU-Sim

User Guide

A browser-based GPU cluster lab: 8 DGX nodes, 64 GPUs,
a realistic terminal, live dashboards and 40 guided labs.

Install · Interface · Labs · Sandbox · Commands · Use cases



01 · OVERVIEW

What MTGPU-Sim is

MTGPU-Sim is a GPU cluster lab that runs entirely in your web browser. It simulates an 8-node DGX cluster (dgx-00 to dgx-07, 8 GPUs each) with InfiniBand networking, a Slurm scheduler, containers and parallel storage. You type real datacenter commands into a terminal and get realistic output that reacts to the state of the cluster. Break something on purpose and every tool shows the symptom.

There are two views, switched from the pill control in the top bar:

- **Simulator**: the live cluster. A dashboard, a terminal, and a Sandbox for injecting faults.
- **Labs**: 40 story-driven missions that walk you through bring-up, fabric, scheduling, validation and troubleshooting, plus on-demand live incidents.

Realistic terminal

nvidia-smi, dcgmi, ipmitool, ibstat, ibdiagnet, mlxlink, sinfo, srun, docker, enroot, lfs, NCCL and HPL tests, and 100+ Linux tools. Includes pipes, history and tab completion.

Live cluster model

Temperature, power, clocks, ECC, NVLink and InfiniBand counters change over time and respond to load, faults and your fixes.

Six architectures

DGX A100, H100, H200, B200, GB200 and VR200. GPU memory, NVLink, network rates, limits and driver versions adapt to the one you pick.

Fault injection

XID errors, ECC, thermal, NVLink, PCIe, power, IB port errors, hangs. They persist until you remediate them.

Guided labs

40 missions with a briefing, step-by-step tasks, knowledge checks and a debrief. Three difficulty tiers and spaced review.

Private and offline-friendly

Nothing to sign up for. Progress is saved in your browser, and the simulation runs on your machine.

02 · SETUP

Install and run

MTGPU-Sim is a Node.js web app. You install it once, start a local server, and open it in your browser.

Platforms

Platform	Supported	Notes
macOS (Apple silicon or Intel)	Yes	Install Node with Homebrew (`brew install node@22`) or the Node installer.
Windows 10 / 11	Yes	Install Node LTS. Run the same commands in PowerShell or Windows Terminal.
Linux (Ubuntu, Debian, Fedora…)	Yes	Install Node 20/22 from your package manager or nvm.
Browsers	Chrome, Edge, Firefox, Safari	Use a recent version. Best on a desktop or laptop screen at least 1280 px wide.
Tablets / phones	Limited	It runs, but the dashboard and terminal are designed for a desktop layout.

Requirements

- **Node.js 20 or 22 (LTS)** with npm. Check with `node -v` and `npm -v`.
- About 1 GB of free disk space for dependencies. No GPU is needed.

First run

```
cd dc-lab-sim-main          # the MTGPU-Sim folder
npm install                 # first time only (and after updates)
npm run dev                 # start the local server
```

Open **http://localhost:5173** in your browser. Keep the terminal window open while you use the sim. Press `Ctrl+C` there to stop it. Next time, only `npm run dev` is needed.

Useful commands

Command	Purpose
`npm run dev`	Start the app at localhost:5173
`npm run build` then `npm run preview`	Build an optimized copy and serve it at localhost:4173
`npm run test:run`	Run the unit tests
`npm run lint`	Check the code style

If something goes wrong
Port busy: Vite picks the next free port; use the URL it prints. Install errors: run `rm -rf node_modules && npm install`. Odd state after an update: hard-refresh the page, or press the reset button in the top bar.

03 · INTERFACE

Find your way around

Top bar

Element	What it does
MTGPU-Sim logo	App name
Simulator / Labs switch	Moves between the live cluster and the lab list. An amber badge on Labs means spaced-review drills are due; click it to start one.
Cluster status	Running indicator, cluster name, and node and GPU counts
Play / Pause	Starts or freezes the simulation clock. Pause to read values that would otherwise keep moving.
Reset	Rebuilds the cluster to a clean, healthy state

Simulator view

Panel	What you'll see and do
Dashboard › Overview	Per-node cards: GPU health, utilization, memory, temperature, power, Slurm state, alerts. Also the architecture selector (A100 → VR200).
Dashboard › NVLink Topology	Interactive GPU and NVSwitch map for the selected node. Degraded links are highlighted.
Dashboard › InfiniBand Fabric	Leaf and spine fabric map with port health. Hover to trace a node's links.
Dashboard › Historical Metrics	Charts of utilization, temperature and power over time.
Terminal	Your shell on the cluster. The prompt shows the current node; `ssh dgx-NN` moves to another. Split the terminal into panes; `Ctrl+\` resets the split.
Sandbox	Inject faults or workloads into a node or GPU, then diagnose from the terminal. **Clear All** / **Reset to healthy** undo them.

Drag the divider between the dashboard and the terminal to resize them. The layout is remembered.

Terminal shortcuts

Keys	Action	Keys	Action
`Tab`	Complete a command or flag	`Ctrl+A` / `Ctrl+E`	Start / end of line
`↑` / `↓`	Command history	`Ctrl+W` / `Ctrl+U` / `Ctrl+K`	Delete word / to start / to end
`Ctrl+R`	Search history	`|`	Pipe output: `dmesg | grep -i xid`
`help <cmd>`	Command reference	`hint`	A nudge during a lab step

04 · LABS

Run a lab

Open **Labs**, pick a mission card, and follow the flow:

1 · Briefing	2 · Steps	3 · Checks	4 · Debrief
Read the story and the goal. Skippable for the two intro labs.	A mission panel shows the current task. Type the commands in the terminal. The step completes when the expected command runs.	Short questions appear inline. Answer them to continue.	A summary of what you fixed and your score. Return to Labs.

- **Abort** leaves a mission at any time. The **Dashboard** button in the mission bar slides the live cluster view in and out.
- Labs run in an **isolated copy** of the cluster. Faults injected by a lab never leak into the Simulator view.
- **Tiers**: 1 Guided (commands shown), 2 Choice (you pick the tools), 3 Realistic (symptom only). Higher tiers unlock as you progress.
- **Live incidents** (bottom of the Labs page) generate a random fault for you to diagnose against a scoring rubric, ending in an after-action review.

All 40 labs

#	Lab	Area	Level	Min
01	Terminal Bootcamp: Getting Around	Foundations	Beginner	12
02	Terminal Bootcamp: Output Filtering	Foundations	Beginner	13
03	First Light: Bringing a DGX Node Online	Bring-up	Beginner	20
04	Driver & Firmware Validation	Bring-up	Beginner	20
05	The BIOS Verification	Bring-up	Beginner	15
06	The Rack Expansion	Bring-up	Beginner	28
07	The Midnight Deployment	Bring-up	Intermediate	25
08	The Firmware Emergency	Bring-up	Intermediate	20
09	The Driver Disaster	Bring-up	Advanced	22
10	Network Bonding Blues	Bring-up	Intermediate	20
11	Wiring the Fabric: InfiniBand Initialization	Bring-up	Intermediate	20
12	NVLink & Fabric Manager Activation	Bring-up	Intermediate	20
13	The Fabric Manager Awakening	Bring-up	Intermediate	20
14	The Fiber Optic Fault	Physical layer	Beginner	15
15	The Fabric Health Check	Physical layer	Intermediate	20
16	The NVLink Mystery	Physical layer	Advanced	25
17	The PCIe Puzzle	Physical layer	Advanced	23
18	The Slurm Setup	Control plane	Beginner	25
19	Standing Up the Scheduler: Slurm Partitions & Health	Control plane	Intermediate	20
20	The Job Scheduler Setup	Control plane	Intermediate	25
21	Container Runtime: Docker, Enroot & Pyxis with NGC	Control plane	Intermediate	22
22	Container Orchestration	Control plane	Intermediate	20
23	The Container Crisis	Control plane	Intermediate	20
24	The Storage Architect	Control plane	Intermediate	20
25	The Storage Showdown	Control plane	Intermediate	22
26	The Health Inspector	Validation	Intermediate	20
27	Validation: DCGM Diagnostic Levels & ECC Baseline	Validation	Advanced	25
28	Validation: Multi-Node NCCL All-Reduce	Validation	Advanced	25
29	The NCCL Championship	Validation	Advanced	25
30	The Bandwidth Bottleneck	Validation	Intermediate	23
31	Fabric Under Pressure	Validation	Advanced	25
32	Linpack Showdown	Validation	Advanced	25

#	Lab	Area	Level	Min
33	The Benchmark Battle	Validation	Advanced	26
34	The NCCL Nightmare	Validation	Advanced	24
35	The Silent Cluster	Validation	Advanced	25
36	The Cluster Certification	Validation	Advanced	30
37	The Memory Mystery	Troubleshooting	Intermediate	22
38	The XID Investigation	Troubleshooting	Advanced	24
39	Cable Detective	Troubleshooting	Advanced	25
40	The Network Nightmare	Troubleshooting	Advanced	25

Suggested order: top to bottom. Start with the two Foundations labs, then bring-up, then the others.

05 · SANDBOX

Practice with faults

The Sandbox is free practice with no scoring. Switch from **Terminal** to **Sandbox** in the Simulator view, choose a node (and a GPU where relevant), inject a fault, then return to the terminal and find it.

Fault	Where it shows up	Try
XID error	Kernel log, nvidia-smi, dashboard alert	`dmesg | grep -i xid`, `nvidia-smi -q`
ECC / severe ECC	ECC counters, row remapping, XID 48/94	`nvidia-smi -q -d ECC`
Thermal alert	Temperatures climb, clocks throttle	`nvidia-smi -q -d PERFORMANCE`, `sensors`
Power issue	Power at cap, throttle reasons	`nvidia-smi -q -d POWER`, `nvidia-smi -pl`
NVLink down	Topology and link status change	`nvidia-smi topo -m`, `nvidia-smi nvlink -s -e`
PCIe error	GPU drops off the bus	`lspci | grep -i nvidia`, `dmesg`
IB port error	Port state, width, error counters	`ibstat`, `iblinkinfo`, `perfquery`
GPU hang	GPU stops responding	`nvidia-smi`, `dcgmi diag -r 1`

Fixing it

- Faults persist until you remediate them. A thermal fault keeps heating the GPU until you lower its power limit, and a GPU past its shutdown limit halts.
- Follow a realistic ladder: capture evidence (`nvidia-bug-report.sh`), drain the node (`scontrol update nodename=dgx-NN state=drain reason='…'`), reset or reseat, validate (`dcgmi diag -r 1`), then resume the node.
- Some faults can't be fixed in software and need a physical action (reseat or RMA) from the Sandbox panel. The sim gates those behind a bug report and a drained node.
- **Clear All** or **Reset to healthy** returns the node to normal. **Reset** in the top bar rebuilds the whole cluster.

> **Drill idea**
> Inject one fault on a random node, close the Sandbox panel, and time how long it takes you to find the node, the GPU and the cause using only the terminal.

06 · COMMANDS

Quick command reference

Area	Commands
GPU	`nvidia-smi`, `-q`, `-q -d ECC|POWER|PERFORMANCE|TEMPERATURE`, `topo -m`, `nvlink -s/-e`, `-pm 1`, `-pl <W>`, `-r -i <n>`, `mig …`, `dmon`, `pmon`
DCGM	`dcgmi discovery -l`, `dcgmi diag -r 1|2|3|4`, `dcgmi health -c`, `dcgmi group -c/-a/-l`, `dcgmi dmon -e …`, `dcgmi stats`, `dcgmi policy`
Node / BMC	`ipmitool sel elist`, `sensor list`, `mc info`, `fru print`, `chassis status`; `dmidecode -t bios|system|memory`; `sensors`; `nvsm show health`
Kernel / OS	`lspci`, `lsmod | grep nvidia`, `dmesg | grep -i nvrm`, `cat /proc/driver/nvidia/version`, `uname -a`, `journalctl -u <svc>`, `systemctl status nvidia-fabricmanager`
InfiniBand	`ibstat`, `iblinkinfo`, `ibnetdiscover`, `ibhosts`, `ibswitches`, `ibdiagnet`, `perfquery [-x]`, `ibporterrors`, `sminfo`, `ibping`, `ibtracert`, `ib_write_bw`
NIC / cables	`mst start|status`, `mlxconfig -d <dev> q`, `mlxlink -d mlx5_0`, `mlxcables`, `mlxfwmanager --query`, `ofed_info -s`
Slurm	`sinfo [-N -l]`, `squeue [-w node]`, `scontrol show node|partition|config`, `scontrol update … state=drain|resume`, `sbatch`, `srun --gres=gpu:N`, `sacct`, `scancel`
Containers	`docker run --gpus all …`, `docker ps`, `enroot import|create|list`, `srun --container-image=…`, `nvidia-container-cli info`
Storage	`lfs df -h`, `lfs check servers`, `lfs getstripe|setstripe`, `nfsstat -c|-m`, `df -hT`, `iostat -x`
Benchmarks	`all_reduce_perf -b 8 -e 256M -g 8`, `mpirun -np 64 all_reduce_perf`, `hpl`, `gpu-burn`, `nvbandwidth`, `clusterkit`

Type `help` for the full list, or `help <command>` for flags and examples.

07 · USE CASES

Ways to use MTGPU-Sim

Learn GPU infrastructure
Work through the 40 labs in order. Each one teaches a tool and the healthy-versus-broken signal to look for.

Interview preparation
Redo labs at Tier 3, say your reasoning aloud, and use live incidents as mock troubleshooting rounds.

Certification practice
Drill the bring-up, validation and troubleshooting workflows that infrastructure certifications test.

Team onboarding
Give new engineers a safe cluster to break. Pair a lab with a short debrief on how your real environment differs.

Runbook rehearsal
Walk through drain → evidence → fix → validate → resume until it's muscle memory, before doing it on production.

Demos and teaching
Switch architectures live to show how A100, H100 and B200 differ in topology, bandwidth and limits.

A simple weekly routine

Day	Do
1–3	Two or three new labs a day. Write one line per lab: symptom → command → abnormal field → fix.
4	Sandbox drills: inject faults you've seen this week and diagnose them blind.
5	Repeat the week's hardest lab at a higher tier and on a different architecture.
6	One live incident. Review the after-action report.

Good to know

- Simulated output is modeled on real tools but approximate. Treat numbers as directionally right, not as benchmarks.
- Timings are compressed: long diagnostics and burn-ins finish in seconds.
- Progress lives in this browser. Clearing site data or using a private window starts fresh.
- Product names used in the simulator are trademarks of their respective owners.